Ethan A. Wideman

Asheville, NC | 22eawideman@gmail.com | (765) 274-9965 | LinkedIn | GitHub | Portfolio

EDUCATION

Indiana University – Luddy School of Informatics – Bloomington, IN **May 2026**
Bachelor of Science in Computer Science - Focus in Software Development

TECHNICAL SKILLS

Languages: Java, JavaScript, TypeScript, Python, C, HTML/CSS
Frameworks/Libraries: React, Next.js, Node.js, Express, Spring Boot, NumPy, Pandas, Matplotlib, PyTorch
Tools/Cloud: PostgreSQL, Redis, AWS (EC2, S3, CloudFront, RDS, Batch, Route53), Docker, Kubernetes, Git, FFmpeg, HLS, TUS

PROJECTS

Character Level GPT Text Generation *(Python, PyTorch)* **Spring 2026**

- Built a **character-level GPT** from scratch in **PyTorch**, implementing **multi-head self-attention**, causal masking, positional encoding, and scaled dot-product attention without high-level abstractions
- Trained on **~5MB** of classic literature, achieving **1.25 training loss** over 50,000 steps and generating stylistically coherent prose with cross-author style blending
- Implemented manual scaled **dot-product attention** with causal masking, learned Q/K/V projections, **8-head attention**, and cosine annealing learning rate scheduling

Neural Collaborative Filtering Recommendation System (*Python, NumPy, Pandas, Matplotlib)* **Spring 2026**

- Implemented a **Neural Collaborative Filtering** (NCF) recommendation system from scratch in **NumPy,** including matrix factorization, MLP forward/backward passes, and the Adam optimizer — no ML frameworks used
- Trained on MovieLens 1M (1 million ratings, 6,000 users) with **temporal leave-one-out splitting** and **negative sampling**, achieving HR@10 of 0.667 and NDCG@10 of 0.402, comparable to published NCF benchmarks
- Engineered a sparse **Adam optimizer** for embedding updates using scatter-add gradient accumulation, improving HR@10 by 47% over a popularity baseline and reducing training loss from 0.693 to 0.196 over 40 epochs
- Developed an interactive demo using **Matplotlib** to visualize how recommendations shift as a user's watch history grows, comparing model output against a popularity baseline across a 100-candidate evaluation set

Streamlair — Full-Stack Video Streaming Platform *(Next.js, Node.js, PostgreSQL, AWS, Kubernetes)* **Spring 2026**
Live Demo: *streamlair.net*

- Developed and deployed a full-stack video streaming platform using **Next.js, Node.js/Express, PostgreSQL, and AWS** supporting upload, transcoding, and playback
- Built scalable video processing using **AWS Batch, EC2 Spot, FFmpeg, and HLS transcoding** (360p–1080p), offloading compute-intensive workloads from API services
- Implemented resumable uploads, adaptive **HLS** playback, and responsive SSR frontend experiences
- Architected backend infrastructure with **Redis/BullMQ queues, Google OAuth**, transactional email, and **S3 + CloudFront** delivery
- Containerized services with **Docker** and validated scalability using **Kubernetes HPA** and **k6** load testing supporting 500 concurrent virtual users

Studymate — Full-Stack LMS Web Application (*React, Next.js, Express, PostgreSQL*) **Spring 2025**

- Collaborated on a team to build a **full-stack learning management system (LMS)** supporting course creation, messaging, and grading workflows
- Developed **RESTful APIs** using **Node.js/Express** and implemented database operations with **PostgreSQL**
- Built responsive frontend components with **React (Next.js)** and **Tailwind CSS** for an intuitive user experience
- Implemented **CI/CD** pipelines with **GitHub Actions** to automate testing and deployment workflows

WORK EXPERIENCE

Wendy's of Bloomington, Bloomington, IN **Nov 2024 - Nov 2025**
Crew Member

- Delivered fast, accurate service in a **high-volume environment serving hundreds of customers daily**
- Maintained **quality and speed** while preparing concurrent orders under time constraints
- Collaborated across roles to **support team efficiency** during peak hours

McDonald's, Elwood, IN **Oct 2020 - July 2022**
Crew Member

- **Trained and onboarded** new employees on POS systems, food safety, and operations
- **Coordinated team workflow** during peak hours to maintain service **speed and accuracy**